

Relational Algebra

Lecture 03

Borough of Manhattan Community College

Introduction

- ❑ The relational algebra is a theoretical language with operations that work on one or more relations to define another relation without changing the original relation(s).
- ❑ The relational algebra has been used as the basis for higher-level Data Manipulation Languages (DMLs) for relational databases.
- ❑ The relational algebra illustrates the basic operations required of any DML and serve as the standard of comparison for other relational languages.

Introduction

- ❑ Expressions can be nested in the relational algebra.
Relations are closed under the algebra, just as numbers are closed under arithmetic operations.
- ❑ The five fundamental operations in relational algebra—Selection, Projection, Cartesian product, Union, and Set difference—perform most of the data retrieval operations that we are interested in.
- ❑ In addition, there are also the Join, Intersection, and Division operations, which can be expressed in terms of the five basic operations.
- ❑ The Selection and Projection operations are **unary** operations, as they operate on one relation. The other operations work on pairs of relations and are therefore called **binary** operations.

Selection

$\sigma_{\text{predicate}}(\mathbf{R})$

The Selection operation works on a single relation R and defines a relation that contains only those tuples of R that satisfy the specified condition (*predicate*).



Selection

Selection

□ Example:

List all staff with a salary greater than £10000.

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

$\sigma_{\text{salary} > 10000}(\text{Staff})$

Selection

List all staff with a salary greater than £10000.

$\sigma_{\text{salary} > 10000}(\text{Staff})$

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003

- ❑ The input relation is **Staff** and the predicate is **salary > 10000**.
- ❑ The Selection operation defines a relation containing only those Staff tuples with a salary greater than £10000.
- ❑ More complex predicates can be generated using the logical operators (AND), (OR), and ~ (NOT).

Projection

$$\Pi_{a_1 \dots, a_n}(\mathbf{R})$$

The Projection operation works on a single relation R and defines a relation that contains a vertical subset of R , extracting the values of specified attributes and eliminating duplicates.



Projection



Selection

Projection

EXAMPLE

Produce a list of all staff, showing only the staffNo, fName, lName, and salary details.

$\Pi_{\text{staffNo, fName, lName, salary}}(\text{Staff})$

staffNo	fName	lName	salary
SL21	John	White	30000
SG37	Ann	Beech	12000
SG14	David	Ford	18000
SA9	Mary	Howe	9000
SG5	Susan	Brand	24000
SL41	Julie	Lee	9000

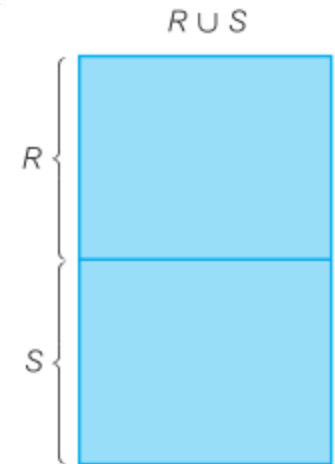
- ❑ The Projection operation defines a relation that contains only the designated Staff attributes **staffNo**, **fName**, **lName**, and **salary**, in the specified order.

Union

$R \cup S$

The union of two relations R and S defines a relation that contains all the tuples of R , or S , or both R and S , duplicate tuples being eliminated. R and S must be union-compatible.

- ❑ If R and S have I and J tuples, respectively, their union is obtained by concatenating them into one relation with a maximum of $(I + J)$ tuples.
- ❑ Union is possible only if the schemas of the two relations match, that is, if they have the same number of attributes with each pair of corresponding attributes having the same domain.
- ❑ In other words, the relations must be **union-compatible**. Note that attributes names are not used in defining union-compatibility.



Union

EXAMPLE

List all cities where there is either a branch office or a property for rent.

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Union

EXAMPLE

List all cities where there is either a branch office or a property for rent.

$$\Pi_{\text{city}}(\text{Branch}) \cup \Pi_{\text{city}}(\text{PropertyForRent})$$

city
London
Aberdeen
Glasgow
Bristol

- ❑ To produce union-compatible relations, we first use the Projection operation to project the Branch and PropertyForRent relations over the attribute city, eliminating duplicates where necessary.
- ❑ We then use the Union operation to combine these new relations to produce the result.

Set Difference

R - S

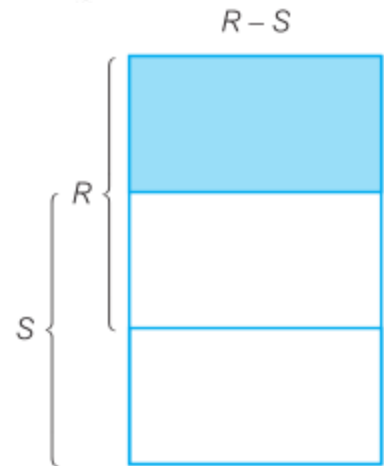
The Set difference operation defines a relation consisting of the tuples that are in relation R, but not in S. R and S must be union-compatible.

EXAMPLE

List all cities where there is a branch office but no properties for rent.

$\Pi_{\text{city}}(\text{Branch}) - \Pi_{\text{city}}(\text{PropertyForRent})$

city
Bristol



- ❑ As in the previous example, we produce union-compatible relations by projecting the Branch and PropertyForRent relations over the attribute city.
- ❑ We then use the Set difference operation to combine these new relations to produce the result.

Intersection

$R \cap S$

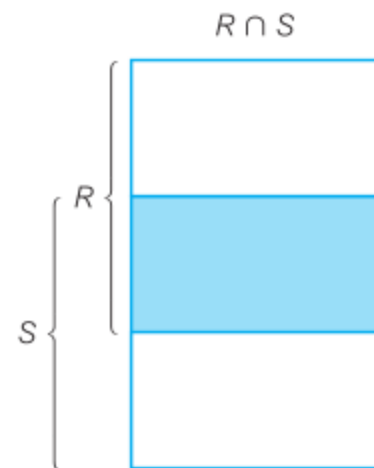
The Intersection operation defines a relation consisting of the set of all tuples that are in both R and S. R and S must be union-compatible.

EXAMPLE

List all cities where there is both branch offices and properties for rent.

$\Pi_{\text{city}}(\text{Branch}) \cap \Pi_{\text{city}}(\text{PropertyForRent})$

city
Aberdeen
London
Glasgow



- ❑ As in the previous example, we produce union-compatible relations by projecting the Branch and PropertyForRent relations over the attribute city.
- ❑ We then use the Intersection operation to combine these new relations to produce the result.

Cartesian Product

$R \times S$

The Cartesian product operation defines a relation that is the concatenation of every tuple of relation R with every tuple of relation S .

- ❑ The Cartesian product operation multiplies two relations to define another relation consisting of all possible pairs of tuples from the two relations.

P		Q	$P \times Q$	
A		B	A	B
a		1	a	1
a		2	a	2
a		3	a	3
b		1	b	1
b		2	b	2
b		3	b	3

- ❑ Therefore, if one relation has I tuples and N attributes and the other has J tuples and M attributes, the Cartesian product relation will contain $(I * J)$ tuples with $(N + M)$ attributes.
- ❑ It is possible that the two relations may have attributes with the same name. In this case, the attribute names are prefixed with the relation name to maintain the uniqueness of attribute names within a relation.

Cartesian Product

EXAMPLE

List the names and comments of all clients who have viewed a property for rent.

Client

clientNo	fName	lName	telNo	prefType	maxRent	eMail
CR76	John	Kay	0207-774-5632	Flat	425	john.kay@gmail.com
CR56	Aline	Stewart	0141-848-1825	Flat	350	astewart@hotmail.com
CR74	Mike	Ritchie	01475-392178	House	750	mritchie01@yahoo.co.uk
CR62	Mary	Tregear	01224-196720	Flat	600	maryt@hotmail.co.uk

Viewing

clientNo	propertyNo	viewDate	comment
CR56	PA14	24-May-13	too small
CR76	PG4	20-Apr-13	too remote
CR56	PG4	26-May-13	
CR62	PA14	14-May-13	no dining room
CR56	PG36	28-Apr-13	

Cartesian Product

EXAMPLE

List the names and comments of all clients who have viewed a property for rent.

- ❑ The names of clients are held in the Client relation and the details of viewings are held in the Viewing relation.
- ❑ To obtain the list of clients and the comments on properties they have viewed, we need to combine these two relations.

$$(\Pi_{\text{clientNo, fName, lName}} (\text{Client})) \times (\Pi_{\text{clientNo, propertyNo, comment}} (\text{Viewing}))$$

Cartesian Product

$$(\Pi_{\text{clientNo, fName, lName}}(\text{Client})) \times (\Pi_{\text{clientNo, propertyNo, comment}}(\text{Viewing}))$$

client.clientNo	fName	lName	Viewing.clientNo	propertyNo	comment
CR76	John	Kay	CR56	PA14	too small
CR76	John	Kay	CR76	PG4	too remote
CR76	John	Kay	CR56	PG4	
CR76	John	Kay	CR62	PA14	no dining room
CR76	John	Kay	CR56	PG36	
CR56	Aline	Stewart	CR56	PA14	too small
CR56	Aline	Stewart	CR76	PG4	too remote
CR56	Aline	Stewart	CR56	PG4	
CR56	Aline	Stewart	CR62	PA14	no dining room
CR56	Aline	Stewart	CR56	PG36	
CR74	Mike	Ritchie	CR56	PA14	too small
CR74	Mike	Ritchie	CR76	PG4	too remote
CR74	Mike	Ritchie	CR56	PG4	
CR74	Mike	Ritchie	CR62	PA14	no dining room
CR74	Mike	Ritchie	CR56	PG36	
CR62	Mary	Tregear	CR56	PA14	too small
CR62	Mary	Tregear	CR76	PG4	too remote
CR62	Mary	Tregear	CR56	PG4	
CR62	Mary	Tregear	CR62	PA14	no dining room
CR62	Mary	Tregear	CR56	PG36	

- ❑ This relation contains more information than we require.
- ❑ To obtain the required list, we need to apply a Selection operation on this relation.

Cartesian Product

The complete operation is thus:

$$\sigma_{\text{Client.clientNo} = \text{Viewing.clientNo}}((\Pi_{\text{clientNo, fName, lName}}(\text{Client})) \times (\Pi_{\text{clientNo, propertyNo, comment}}(\text{Viewing})))$$

client.clientNo	fName	lName	Viewing.clientNo	propertyNo	comment
CR76	John	Kay	CR76	PG4	too remote
CR56	Aline	Stewart	CR56	PA14	too small
CR56	Aline	Stewart	CR56	PG4	
CR56	Aline	Stewart	CR56	PG36	
CR62	Mary	Tregear	CR62	PA14	no dining room

Restricted Cartesian product of reduced Client and Viewing relations.

Join Operations

- ❑ The **Join operation**, which combines two relations to form a new relation, is one of the essential operations in the relational algebra.

- ❑ There are various forms of the Join operation; each with subtle differences:
 - Theta join
 - Equijoin (a particular type of Theta join)
 - Natural join
 - Outer join

Theta join (θ -join)

$$R \bowtie_F S$$

The Theta join operation defines a relation that contains tuples satisfying the predicate F from the Cartesian product of R and S . The predicate F is of the form $R.a_i \theta S.b_j$, where θ may be one of the comparison operators ($<$, $\leq$, $>$, $\geq$, $=$, $\neq$).

- We can rewrite the Theta join in terms of the basic Selection and Cartesian product operations:

$$R \bowtie_F S = \sigma_F(R \times S)$$

- As with a Cartesian product, the degree of a Theta join is the sum of the degrees of the operand relations R and S .
- In the case where the predicate F contains only equality ($=$), the term **Equijoin** is used instead.

Equijoin Operation

EXAMPLE

List the names and comments of all clients who have viewed a property for rent.

- In a previous example we used the Cartesian product and Selection operations to obtain this list.

$$\sigma_{\text{Client.clientNo} = \text{Viewing.clientNo}}((\Pi_{\text{clientNo, fName, lName}}(\text{Client})) \times (\Pi_{\text{clientNo, propertyNo, comment}}(\text{Viewing})))$$

- However, the same result is obtained using the Equijoin operation:

$$(\Pi_{\text{clientNo, fName, lName}}(\text{Client})) \bowtie_{\text{Client.clientNo} = \text{Viewing.clientNo}} (\Pi_{\text{clientNo, propertyNo, comment}}(\text{Viewing}))$$

client.clientNo	fName	lName	Viewing.clientNo	propertyNo	comment
CR76	John	Kay	CR76	PG4	too remote
CR56	Aline	Stewart	CR56	PA14	too small
CR56	Aline	Stewart	CR56	PG4	
CR56	Aline	Stewart	CR56	PG36	
CR62	Mary	Tregear	CR62	PA14	no dining room

Natural Join

$R \bowtie S$

The Natural join is an Equijoin of the two relations R and S over all common attributes x . One occurrence of each common attribute is eliminated from the result.

T		U		$T \bowtie U$		
A	B	B	C	A	B	C
a	1	1	x	a	1	x
b	2	1	y	a	1	y
		3	z			

- ❑ The Natural join operation performs an Equijoin over all the attributes in the two relations that have the same name.
- ❑ The degree of a Natural join is the sum of the degrees of the relations R and S less the number of attributes in x .

Natural Join

EXAMPLE

List the names and comments of all clients who have viewed a property for rent.

- ❑ In the previous example we used the Equijoin to produce this list, but the resulting relation had two occurrences of the join attribute clientNo.
- ❑ We can use the Natural join to remove one occurrence of the clientNo attribute:

$$(\Pi_{\text{clientNo, fName, lName}}(\text{Client})) \bowtie (\Pi_{\text{clientNo, propertyNo, comment}}(\text{Viewing}))$$

clientNo	fName	lName	propertyNo	comment
CR76	John	Kay	PG4	too remote
CR56	Aline	Stewart	PA14	too small
CR56	Aline	Stewart	PG4	
CR56	Aline	Stewart	PG36	
CR62	Mary	Tregear	PA14	no dining room

Outer Join

- ❑ Often in joining two relations, a tuple in one relation does not have a matching tuple in the other relation.
- ❑ We may want tuples from one of the relations to appear in the result even when there are no matching values in the other relation. This may be accomplished using the **Outer join**.

$R \bowtie S$

The (left) Outer join is a join in which tuples from R that do not have matching values in the common attributes of S are also included in the result relation. Missing values in the second relation are set to null.

T		U		$T \bowtie U$		
A	B	B	C	A	B	C
a	1	1	x	a	1	x
b	2	1	y	a	1	y
		3	z	b	2	null

Left Outer join

Outer Join

EXAMPLE:

Produce a status report on property viewings. The report must include all properties.

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Viewing

clientNo	propertyNo	viewDate	comment
CR56	PA14	24-May-13	too small
CR76	PG4	20-Apr-13	too remote
CR56	PG4	26-May-13	
CR62	PA14	14-May-13	no dining room
CR56	PG36	28-Apr-13	

Outer Join

- ❑ In this example, we want to produce a relation consisting of the properties that have been viewed with comments and those that have not been viewed.
- ❑ This can be achieved using the following Outer join:

$(\Pi_{\text{propertyNo, street, city}}(\text{PropertyForRent})) \bowtie \text{Viewing}$

propertyNo	street	city	clientNo	viewDate	comment
PA14	16 Holhead	Aberdeen	CR56	24-May-13	too small
PA14	16 Holhead	Aberdeen	CR62	14-May-13	no dining room
PL94	6 Argyll St	London	null	null	null
PG4	6 Lawrence St	Glasgow	CR76	20-Apr-13	too remote
PG4	6 Lawrence St	Glasgow	CR56	26-May-13	
PG36	2 Manor Rd	Glasgow	CR56	28-Apr-13	
PG21	18 Dale Rd	Glasgow	null	null	null
PG16	5 Novar Dr	Glasgow	null	null	null

Outer Join

- ❑ Strictly speaking, the previous example is a **Left (natural) Outer join**, as it keeps every tuple in the left-hand relation in the result.
- ❑ Similarly, there is a **Right Outer join** that keeps every tuple in the right-hand relation in the result.
- ❑ There is also a **Full Outer join** that keeps all tuples in both relations, padding tuples with nulls when no matching tuples are found.

Division Operation

- Assume that relation R is defined over the attribute set A and relation S is defined over the attribute set B such that B is a subset of A .
- Let $C = A - B$, that is, C is the set of attributes of R that are not attributes of S . We have the following definition of the Division operation.

$$R \div S$$

The Division operation defines a relation over the attributes C that consists of the set of tuples from R that match the combination of **every** tuple in S .



Division (shaded area)

R		S	$R \div S$
A	B	B	A
a	1	1	a
a	2	2	b
b	1		
b	2		
c	1		

Example of division

Division Operation

EXAMPLE

Identify all clients who have viewed all properties with three rooms.

- We can use the Selection operation to find all properties with three rooms followed by the Projection operation to produce their property numbers.

$$(\Pi_{\text{clientNo, propertyNo}}(\text{Viewing})) \div (\Pi_{\text{propertyNo}}(\sigma_{\text{rooms} = 3}(\text{PropertyForRent})))$$

$\Pi_{\text{clientNo, propertyNo}}(\text{Viewing})$

clientNo	propertyNo
CR56	PA14
CR76	PG4
CR56	PG4
CR62	PA14
CR56	PG36

$\Pi_{\text{propertyNo}}(\sigma_{\text{rooms}=3}(\text{PropertyForRent}))$

propertyNo
PG4
PG36

RESULT

clientNo
CR56

Division Operation

R		
A	B	C
a	4	x
a	4	y
b	3	z
c	3	z
b	4	z
a	3	z
c	4	x

S	
B	C
3	z
4	x

$R \div S$	
A	
a	
c	

Rename

- ❑ We can use the Rename operation ρ (rho) to give a name to the result of a relational algebra operation.
- ❑ Rename allows an optional name for each of the attributes of the new relation to be specified.

$\rho_S(E)$ or
 $\rho_{S(a_1, a_2, \dots, a_n)}(E)$

The Rename operation provides a new name S for the expression E , and optionally names the attributes as $a_1, a_2, \dots, a_n$.

Aggregation and Grouping Operations

- ❑ As well as simply retrieving certain tuples and attributes of one or more relations, we often want to perform some form of additional operations such as:
 - Summation or aggregation of data, similar to the totals at the bottom of a report.
 - Grouping of data, similar to subtotals in a report.
- ❑ These operations cannot be performed using the basic relational algebra operations considered earlier.
- ❑ However, additional operations , namely **Aggregate** and **Grouping operations**, have been proposed.

Aggregate Operations

$AL(R)$

Applies the aggregate function list, AL, to the relation R to define a relation over the aggregate list. AL contains one or more (<aggregate_function>, <attribute>) pairs.

The main aggregate functions are:

- COUNT – returns the number of values in the associated attribute.
- SUM – returns the sum of the values in the associated attribute.
- AVERAGE – returns the average of the values in the associated attribute.
- MIN – returns the smallest value in the associated attribute.
- MAX – returns the largest value in the associated attribute.

Aggregate Operations

EXAMPLE

How many properties cost more than £350 per month to rent?

- We can use the aggregate function COUNT to produce the relation R:

$\text{COUNT propertyNo } (\sigma_{\text{rent} > 350} (\text{PropertyForRent}))$

Or

$\rho_R(\text{myCount}) \text{ COUNT propertyNo } (\sigma_{\text{rent} > 350} (\text{PropertyForRent}))$

myCount
5

Aggregate Operations

EXAMPLE

Find the minimum, maximum, and average staff salary.

- We can use the aggregate functions - MIN, MAX, and AVERAGE - to produce the relation R:

MIN salary, MAX salary, AVERAGE salary (Staff)

Or

$\rho_R(\text{myMin, myMax, myAverage})$ MIN salary, MAX salary, AVERAGE salary (Staff)

myMin	myMax	myAverage
9000	30000	17000

Grouping Operations

GA AL(R)

Groups the tuples of relation R by the grouping attributes, GA, and then applies the aggregate function list AL to define a new relation. AL contains one or more (<aggregate_function>, <attribute>) pairs. The resulting relation contains the grouping attributes, GA, along with the results of each of the aggregate functions.

The general form of the grouping operation is as follows:

$$a_1, a_2, \dots, a_n \text{ } \langle A_p a_p \rangle, \langle A_q a_q \rangle, \dots, \langle A_z a_z \rangle (R)$$

where R is any relation, $a_1, a_2, \dots, a_n$ are attributes of R on which to group, $a_p, a_q, \dots, a_z$ are other attributes of R, and $A_p, A_q, \dots, A_z$ are aggregate functions.

The tuples of R are partitioned into groups such that:

- all tuples in a group have the same value for $a_1, a_2, \dots, a_n$;
- tuples in different groups have different values for $a_1, a_2, \dots, a_n$.

Grouping Operations

EXAMPLE

Find the number of staff working in each branch and the sum of their salaries.

- We first need to group tuples according to the branch number and then use the aggregate functions COUNT and SUM to produce the required relation.

branchNo COUNT staffNo, SUM salary (Staff)

Or

$\rho_R(\text{branchNo, myCount, mySum})$ branchNo COUNT staffNo, SUM salary (Staff)

branchNo	myCount	mySum
B003	3	54000
B005	2	39000
B007	1	9000



Examples of Relational Algebra Queries

Library

libraryNo	address	city	zipCode
L001	1250 Broadway	New York	10001
L002	360 Hudson ST	New York	10013
L003	1250 Broadway	Boston	22116

User

userNO	name	age	libraryNo
U001	Colin Wilson	17	L001
U002	John Parker	35	L001
U003	Elizabeth Sagan	16	L002
U004	Matthew Arnold	23	L003

Book

bookNo	title	author	year	price	available	libraryNo
B001	A Darkling Plain	Philip Reeve	2006	70	Y	L001
B002	Enemy Pie	Derek Munson	2000	65	N	L003
B003	Postern of Fate	Agatha Christie	1973	85	N	L001
B004	The Last Enemy	Richard Hillary	1944	65	Y	L001
B005	Endless Night	Agatha Christie	1967	49	Y	L002
B006	The Millstone	Margaret Drabble	1965	55	Y	L003

BookLoan

bookNo	dateOut	dateDue	userNo
B003	10-Jan-17	24-Jan-17	U002
B005	22-Jan-17	10-Feb-17	U003
B004	27-Jan-17	15-Feb-17	U002
B003	9-Feb-17	23-Feb-17	U004
B005	13-Feb-17	27-Feb-17	U002
B006	1-Mar-17	15-Mar-17	U004

1. List all book titles.

$\Pi_{title}(Book)$

2. List full details of all books that are currently available.

$\sigma_{available='Y'}(Book)$

3. List the name and age of all users under age 18.

$\Pi_{name,age}(\sigma_{age<18}(User))$

4. List the title, the year and the availability of all books published after 1960.

$\Pi_{title,year,available}(\sigma_{year>1960}(Book))$

Library

libraryNo	address	city	zipCode
L001	1250 Broadway	New York	10001
L002	360 Hudson ST	New York	10013
L003	1250 Broadway	Boston	22116

Book

bookNo	title	author	year	price	available	libraryNo
B001	A Darkling Plain	Philip Reeve	2006	70	Y	L001
B002	Enemy Pie	Derek Munson	2000	65	N	L003
B003	Postern of Fate	Agatha Christie	1973	85	N	L001
B004	The Last Enemy	Richard Hillary	1944	65	Y	L001
B005	Endless Night	Agatha Christie	1967	49	Y	L002
B006	The Millstone	Margaret Drabble	1965	55	Y	L003

User

userNO	name	age	libraryNo
U001	Colin Wilson	17	L001
U002	John Parker	35	L001
U003	Elizabeth Sagan	16	L002
U004	Matthew Arnold	23	L003

BookLoan

bookNo	dateOut	dateDue	userNo
B003	10-Jan-17	24-Jan-17	U002
B005	22-Jan-17	10-Feb-17	U003
B004	27-Jan-17	15-Feb-17	U002
B003	9-Feb-17	23-Feb-17	U004
B005	13-Feb-17	27-Feb-17	U002
B006	1-Mar-17	15-Mar-17	U004

5. List the title, the author, the libraryNo and the library address of all books.

$$\Pi_{\text{title,author,libraryNo}}(\text{Book}) \bowtie \Pi_{\text{libraryNo,address}}(\text{Library})$$

$$\Pi_{\text{title,author,libraryNo,address}}(\text{Book} \bowtie \text{Library})$$

6. List the name and the libraryNo of users from New York libraries, as well as the address of the library.

$$\Pi_{\text{name,libraryNo}}(\text{User}) \bowtie \Pi_{\text{libraryNo,address}}(\sigma_{\text{city}='NewYork'}(\text{Library}))$$

$$\Pi_{\text{name,libraryNo,address}}(\text{User} \bowtie \sigma_{\text{city}='NewYork'}(\text{Library}))$$

$$\Pi_{\text{name,libraryNo,address}}(\sigma_{\text{city}='NewYork'}(\text{User} \bowtie \text{Library}))$$

Library

libraryNo	address	city	zipCode
L001	1250 Broadway	New York	10001
L002	360 Hudson ST	New York	10013
L003	1250 Broadway	Boston	22116

User

userNO	name	age	libraryNo
U001	Colin Wilson	17	L001
U002	John Parker	35	L001
U003	Elizabeth Sagan	16	L002
U004	Matthew Arnold	23	L003

Book

bookNo	title	author	year	price	available	libraryNo
B001	A Darkling Plain	Philip Reeve	2006	70	Y	L001
B002	Enemy Pie	Derek Munson	2000	65	N	L003
B003	Postern of Fate	Agatha Christie	1973	85	N	L001
B004	The Last Enemy	Richard Hillary	1944	65	Y	L001
B005	Endless Night	Agatha Christie	1967	49	Y	L002
B006	The Millstone	Margaret Drabble	1965	55	Y	L003

BookLoan

bookNo	dateOut	dateDue	userNo
B003	10-Jan-17	24-Jan-17	U002
B005	22-Jan-17	10-Feb-17	U003
B004	27-Jan-17	15-Feb-17	U002
B003	9-Feb-17	23-Feb-17	U004
B005	13-Feb-17	27-Feb-17	U002
B006	1-Mar-17	15-Mar-17	U004

7. Display the number of libraries.

COUNT libraryNo (Library)

8. Display the number of books that are currently available.

COUNT bookNo ($\sigma_{available='Y'}$ (Book))

9. Display the number and the total price of books written by the author 'Agatha Christie'.

COUNT bookNo, SUM price ($\sigma_{author='Agatha Christie'}$ (Book))

Library

libraryNo	address	city	zipCode
L001	1250 Broadway	New York	10001
L002	360 Hudson ST	New York	10013
L003	1250 Broadway	Boston	22116

User

userNO	name	age	libraryNo
U001	Colin Wilson	17	L001
U002	John Parker	35	L001
U003	Elizabeth Sagan	16	L002
U004	Matthew Arnold	23	L003

Book

bookNo	title	author	year	price	available	libraryNo
B001	A Darkling Plain	Philip Reeve	2006	70	Y	L001
B002	Enemy Pie	Derek Munson	2000	65	N	L003
B003	Postern of Fate	Agatha Christie	1973	85	N	L001
B004	The Last Enemy	Richard Hillary	1944	65	Y	L001
B005	Endless Night	Agatha Christie	1967	49	Y	L002
B006	The Millstone	Margaret Drabble	1965	55	Y	L003

BookLoan

bookNo	dateOut	dateDue	userNo
B003	10-Jan-17	24-Jan-17	U002
B005	22-Jan-17	10-Feb-17	U003
B004	27-Jan-17	15-Feb-17	U002
B003	9-Feb-17	23-Feb-17	U004
B005	13-Feb-17	27-Feb-17	U002
B006	1-Mar-17	15-Mar-17	U004

10. Display the number of libraries in each city.

city COUNT libraryNo (Library)

11. Display the number and the average age of users in each library.

libraryNo COUNT userNo, AVERAGE age (User)

12. For each author, Display the number and the average price of books published after 1960.

author COUNT bookNo, AVERAGE price ($\sigma_{year > 1960}$ (Book))

Library

libraryNo	address	city	zipCode
L001	1250 Broadway	New York	10001
L002	360 Hudson ST	New York	10013
L003	1250 Broadway	Boston	22116

User

userNO	name	age	libraryNo
U001	Colin Wilson	17	L001
U002	John Parker	35	L001
U003	Elizabeth Sagan	16	L002
U004	Matthew Arnold	23	L003

Book

bookNo	title	author	year	price	available	libraryNo
B001	A Darkling Plain	Philip Reeve	2006	70	Y	L001
B002	Enemy Pie	Derek Munson	2000	65	N	L003
B003	Postern of Fate	Agatha Christie	1973	85	N	L001
B004	The Last Enemy	Richard Hillary	1944	65	Y	L001
B005	Endless Night	Agatha Christie	1967	49	Y	L002
B006	The Millstone	Margaret Drabble	1965	55	Y	L003

BookLoan

bookNo	dateOut	dateDue	userNo
B003	10-Jan-17	24-Jan-17	U002
B005	22-Jan-17	10-Feb-17	U003
B004	27-Jan-17	15-Feb-17	U002
B003	9-Feb-17	23-Feb-17	U004
B005	13-Feb-17	27-Feb-17	U002
B006	1-Mar-17	15-Mar-17	U004

The following relation lists the number of books, the lowest book price, and the highest book price in each library.

col1	col2	col3	col4
L001	3	65	85
L002	1	49	49
L003	2	55	65

13. What is the relational algebra expression that outputs the above relation?

libraryNo *COUNT* *bookNo*, *MIN* *price*, *MAX* *price* (*Book*)

Library

libraryNo	address	city	zipCode
L001	1250 Broadway	New York	10001
L002	360 Hudson ST	New York	10013
L003	1250 Broadway	Boston	22116

User

userNO	name	age	libraryNo
U001	Colin Wilson	17	L001
U002	John Parker	35	L001
U003	Elizabeth Sagan	16	L002
U004	Matthew Arnold	23	L003

Book

bookNo	title	author	year	price	available	libraryNo
B001	A Darkling Plain	Philip Reeve	2006	70	Y	L001
B002	Enemy Pie	Derek Munson	2000	65	N	L003
B003	Postern of Fate	Agatha Christie	1973	85	N	L001
B004	The Last Enemy	Richard Hillary	1944	65	Y	L001
B005	Endless Night	Agatha Christie	1967	49	Y	L002
B006	The Millstone	Margaret Drabble	1965	55	Y	L003

BookLoan

bookNo	dateOut	dateDue	userNo
B003	10-Jan-17	24-Jan-17	U002
B005	22-Jan-17	10-Feb-17	U003
B004	27-Jan-17	15-Feb-17	U002
B003	9-Feb-17	23-Feb-17	U004
B005	13-Feb-17	27-Feb-17	U002
B006	1-Mar-17	15-Mar-17	U004

Let V and W be defined as follows:

$$V = \Pi_{userNo, bookNo} (BookLoan)$$

$$W = \Pi_{bookNo} (\sigma_{author='Agatha Christie'} (Book))$$

14. List the content of the expressions V , W and $V \div W$.

V		W		$V \div W$	
userNo	bookNo	bookNo		userNo	
U002	B003	B003		U002	
U003	B005	B005			
U002	B004				
U004	B003				
U002	B005				
U004	B006				

Library

libraryNo	address	city	zipCode
L001	1250 Broadway	New York	10001
L002	360 Hudson ST	New York	10013
L003	1250 Broadway	Boston	22116

User

userNO	name	age	libraryNo
U001	Colin Wilson	17	L001
U002	John Parker	35	L001
U003	Elizabeth Sagan	16	L002
U004	Matthew Arnold	23	L003

Book

bookNo	title	author	year	price	available	libraryNo
B001	A Darkling Plain	Philip Reeve	2006	70	Y	L001
B002	Enemy Pie	Derek Munson	2000	65	N	L003
B003	Postern of Fate	Agatha Christie	1973	85	N	L001
B004	The Last Enemy	Richard Hillary	1944	65	Y	L001
B005	Endless Night	Agatha Christie	1967	49	Y	L002
B006	The Millstone	Margaret Drabble	1965	55	Y	L003

BookLoan

bookNo	dateOut	dateDue	userNo
B003	10-Jan-17	24-Jan-17	U002
B005	22-Jan-17	10-Feb-17	U003
B004	27-Jan-17	15-Feb-17	U002
B003	9-Feb-17	23-Feb-17	U004
B005	13-Feb-17	27-Feb-17	U002
B006	1-Mar-17	15-Mar-17	U004

15. List the name and age of users who rented any book written by the author 'Agatha Christie'.

$$\Pi_{name,age} (User \bowtie (BookLoan \bowtie \sigma_{author='AgathaChristie'} (Book)))$$

$$\Pi_{name,age} (User \bowtie (\sigma_{author='AgathaChristie'} (BookLoan \bowtie Book)))$$

$$\Pi_{name,age} (\sigma_{author='AgathaChristie'} (User \bowtie (BookLoan \bowtie Book)))$$